

Saurabh Singh

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Objective

Motivated Computer Science Engineering student with strong Full-Stack development skills (**React, JavaScript, Python**). Passionate about building high-performance user interfaces and scalable **backends**. Eager to apply **frontend** expertise to build growth engines and dashboards at **SuperKalam**, with the unique ability to understand and integrate with AI-driven backend systems

Education

Polaris School Of technology

Bachelor of Technology in Computer Science Engineering(AI & ML)

Bangalore, Karanataka, India

Expected Graduation: May 2028

Skills

- **Web Development** React.js,Next.js JavaScript (ES6+), HTML5/CSS3,Tailwind CSS, Node.js, FastAPI
- **Data Science & AI:**Model Evaluation (RMSE/ROUGE), Exploratory Data Analysis (EDA), Collaborative Filtering, NLP, RLHF Concepts, Data Cleaning, Hypothesis Testing.
- **Languages/Framework:**Python (Pandas, NumPy, Scikit-Learn, Surprise), SQL, Java, JavaScript, Node.js, Fast API, HTML/CSS, React
- **Databases:** MySQL, MongoDB
- **Tools & Technologies:** Git, Jupyter Notebooks, Streamlit, HuggingFace, VS Code.

Projects

Citicore [\[Live Demo\]](#)

React,Three.js, D3.js, Framer motion,JavaScript

- Built and deployed a full-stack Smart City Dashboard processing real-time data from **5 APIs** to monitor weather, traffic, air quality, energy, and waste management for **100+ cities** worldwide.
- Implemented **robust API** integration layer with custom service architecture, error handling, and intelligent fallback mechanisms, ensuring consistent data delivery despite rate limits and CORS restrictions
- Implemented advanced data visualization with interactive **charts and 3D animations using Three.js, Plotly, and D3.js**, improving user engagement and data comprehension through intuitive glassmorphism UI design

MovieLens Recommendation System [\[GitHub\]](#)

Python, Scikit-Learn, Surprise Library, SVD, Pandas.

- Built an end-to-end recommendation pipeline comparing **Memory-Based(kNN)** and **Model-Based (SVD)** approaches to solve the 'Choice Paralysis' problem..
- Conducted rigorous model evaluation using RMSE metrics, achieving an 11% accuracy improvement (0.85 vs. 0.95) by optimizing Latent Factors.
- Performed extensive **Data Cleaning and EDA** on 100k+ ratings to filter statistical noise (Long Tail distribution) and address 98% matrix sparsity.
- Analyzed and ranked model predictions to ensure reasoning consistency and relevance for user queries.

LLM-Eval: Comparative Summarization Tool [\[GitHub\]](#)

Python, HuggingFace Transformers, Streamlit, ROUGE Metrics.

- Developed a generic **LLM Evaluation Interface** comparing summarization outputs from **T5** and **BART** models side-by-side.
- Implemented ROUGE scores for automated quality metrics and a **Human-in-the-loop (HITL)** feedback mechanism to manually rank response quality..
- Simulated an **RLHF workflow** by assessing model hallucinations and factual consistency across different text input

Awards & Achievements

- **Runner-Up** in College Hackathon 2025 affiliated with Google for Developers for building a Women Safety web app.